# Code Explanation

**Configuration File for E2E Address Change YUYU**

This is a Python configuration file used for an E2E address change YUYU application. The file contains database connection parameters, Oracle table names, output file paths, and schema definitions for various tables.

## **Overview of the Code**

The configuration file is used to store various parameters and schema definitions required for the E2E address change YUYU application. It includes database connection details, table names, output file paths, and schema definitions for different tables.

## Step 1: Importing Necessary Modules

The code starts by importing the necessary Python modules. The `os` module is used to access environment variables, and the `pyspark.sql.types` module is used to define the schema for various tables.

```
import os
from pyspark.sql.types import StructType, StructField, StringType, T
imestampType, IntegerType
```

## Step 2: Defining Database Connection Parameters

The next step is to define the database connection parameters. The `JDBC_URL`, `DB_USER`, and `DB_PASSWORD` variables are used to store the JDBC URL, database username, and password, respectively. These values are retrieved from environment variables, and default values are provided if the environment variables are not set.

```
JDBC_URL = os.environ.get("JDBC_URL", "jdbc:oracle:thin:@//oracle-ho
st:1521/ZSYSE2EDEV")
DB_USER = os.environ.get("DB_USER", "e2e_user")
DB_PASSWORD = os.environ.get("DB_PASSWORD", "")
PROCESS_USERID = os.environ.get("PROCESS_USERID", "E2E_BATCH")
```

## Step 3: Defining Oracle Table Names

The code then defines the names of the Oracle tables used in the application. The `SOURCE_TABLE`, `YUYU_CLNT_TABLE`, and `YUYUK_CLN_TABLE` variables store the names of the source table and two other tables used in the application.

```
SOURCE_TABLE = "STG_E2E_AC_TXDBH_DATA"
YUYU_CLNT_TABLE = "T_YUYU_CLNT"
YUYUK_CLN_TABLE = "T_YUYUK_CLN"
```

## Step 4: Defining Output File Path

The output file path is defined using the `OUTPUT_FILE_PATH` variable. This variable stores the path where the output file will be written.

```
OUTPUT_FILE_PATH = "/dbfs/mnt/e2e_policy_services/output/DPAddressChangeYUYU"
```

## Step 5: Defining Schema Definitions
The code defines schema definitions for various tables using the `StructType` and `StructField` classes from the `pyspark.sql.types` module. The schema definitions include the column names, data types, and whether the columns are nullable.

```
STG_E2E_AC_TXDBH_DATA_SCHEMA = StructType([
    StructField("T_TX_REQUEST_REQUEST_ID", StringType(), False),
    StructField("T_TX_BASIC_TRANS_ID", StringType(), False),
    # ... other fields ...
])

YUYU_CLNT_SCHEMA = StructType([
    StructField("POL_NO", StringType(), True),
    StructField("POWN_LNM", StringType(), True),
    StructField("POWN_FNM", StringType(), True)
])

YUYUK_CLN_SCHEMA = StructType([
    StructField("POL_NO", StringType(), True),
    StructField("POWN_KNM", StringType(), True)
])

DP_ADDRESS_CHANGE_YUYU_SCHEMA = StructType([
    StructField("SEQUENCE_NUMBER", IntegerType(), False),
    StructField("ORIGIN_REQUEST_ID", StringType(), True),
    # ... other fields ...
])
```

## Step 6: Defining JDBC Connection Properties
Finally, the code defines the JDBC connection properties using a dictionary. The `JDBC_PROPERTIES` dictionary includes the database username, password, and driver class name.

```
JDBC_PROPERTIES = {
    "user": DB_USER,
    "password": DB_PASSWORD,
    "driver": "oracle.jdbc.driver.OracleDriver"
}
```